

Guidelines for Computer Science Thesis Title

Introduction

A Computer Science Thesis, based on *CMO No. 25, s. 2015*, is a technical report on a systematic investigation of a problem that can be solved using Computing. It may include a solution, an approximate or partial solution, a scientific investigation, or the development of results leading to the solution of the problem.

A computer science thesis must be anchored on Computer Science principles.

1. Group Composition

- 1.1. Each group must be composed of students under the same specialization (*e.g., Software Engineering, Data Science, or Artificial Intelligence*).
- 1.2. A group shall have a minimum of three **(3)** and a maximum of four **(4)** members as mentioned during course orientation and reflected in syllabus.
- 1.3. The formation of groups must be finalized and approved by the Thesis Course Adviser before title proposal submission.

2. Thesis Title Requirements

- 2.1. The thesis title must be concise and specific, consisting of no more than sixteen **(16)** words.
- 2.2. The title must indicate the target algorithm, model, or technique to be used in the study (*e.g., "using Random Forest," "through Genetic Algorithm," "based on Convolutional Neural Network"*).
- 2.3. Titles must reflect the main theme and application domain of the research.

3. Research Areas Based on Specialization

Thesis topics shall align with the students' specialization track:

Software Engineering (SE)		
<i>Traditional Image Processing (Non-AI)</i> <i>These rely on mathematical algorithms, not machine learning.</i>		
No.	Technique	Description
1	Histogram Equalization	Enhances image contrast by spreading out pixel intensity values.
2	Canny Edge Detection	Detects edges based on gradients and thresholds.
3	Gaussian Blurring	Reduces noise or smooths images using a Gaussian kernel.
4	Thresholding (Otsu's Method)	Convert grayscale to binary by finding an optimal threshold value.
5	Morphological Operations (Erosion/Dilation)	Removes noise or fills gaps in binary images.

6	Fourier Transform Filtering	Analyzes image frequency components for filtering.
7	Sobel or Prewitt Edge Filters	Detects image gradients using convolution masks.
8	Image Segmentation (Watershed Algorithm)	Divides an image into regions based on pixel gradients.
9	Image Registration (Geometric Transformation)	Aligns two or more images using rotation, scaling, or translation.
10	Color Space Conversion (RGB → HSV/Grayscale)	Convert images between color models for further processing.
11	System Modeling	Focuses on creating abstract representations of software systems to understand, analyze, and design their structure and behavior.
12	Frameworks	Involves developing reusable software structures or platforms that simplify and accelerate application development.
13	Software Design Patterns	Studies reusable solutions to common software design problems to improve code maintainability and scalability.
14	Human-Computer Interaction (HCI)	Explores how users interact with software systems to design intuitive and user-friendly interfaces.
15	Process Automation	Aims to enhance efficiency by automating repetitive software development or business processes using technology.
16	Software Quality Assurance (SQA)	Ensures software reliability and performance through systematic testing, validation, and process improvement.
17	Edge Computing	Design and optimize software systems that process data closer to the source (e.g., IoT devices) instead of relying solely on cloud servers.
18	Blockchain	Developing decentralized and secure software systems using blockchain technology.
19	Model-Driven Software Engineering (MDSE)	Uses models and abstractions to automate software design, generation, and validation processes.
20	Cloud Computing and Service-Oriented Architecture (SOA)	Explores software design and deployment in distributed cloud environments for scalability and flexibility.
21	Expert Systems & Decision Support Systems (DSS)	Expert Systems & DSS are treated as application-oriented projects designed to

		provide automated recommendations or decisions. The emphasis is on developing functional software that supports organizations, businesses, or communities using rule-based engines or structured logic.
22	Game Development	Game Development aligns with software design, visualization, and interaction systems. The focus is on algorithms, performance optimization, rendering pipelines, and interactive interfaces.
23	Image Processing	Covers traditional (non-AI) image manipulation techniques such as filtering, segmentation, edge detection, and transformation. The focus is on mathematical operations and algorithmic processing.
24	Bioinformatics	Covers algorithmic analysis of biological data using computational tools. Includes sequence alignment, genomic analysis, protein structure prediction, and molecular data processing.
25	Graphics Applications	Relates to rendering pipelines, visualization algorithms, geometric modeling, animations, and graphics processing tools. Focus is on mathematical and computational methods in graphics.
26	Parallel Computing	Parallel Computing is a computing approach where multiple tasks are processed at the same time using two or more processors or CPU/GPU cores. Instead of working step-by-step, the main task is divided into smaller subtasks , which are executed simultaneously , allowing the system to finish the work faster and more efficiently .
27	Embedded Computing	Refers to computer systems that are built into larger devices to perform specific, dedicated functions often with real-time constraints.
28	Code Theory	It focuses on detecting and correcting errors that occur when data is sent over noisy channels, such as communication networks, storage devices, or digital systems.
29	Programming Language	Advances in this field enable the creation of safer, more efficient, and more expressive languages , supporting modern applications in software engineering, artificial intelligence, parallel computing, and embedded systems .
30	Mobile Computing	It enables users to remain connected and perform computing tasks without being

		restricted to a fixed location , relying on wireless communication technologies like Wi-Fi, 4G/5G, Bluetooth, and satellite networks .
31	Computer Architecture	It serves as the foundation for building efficient, reliable, and high-performance computing systems , from personal computers to supercomputers.
32	System Software	Research in system software focuses on improving system performance, reliability, security, and scalability . Key areas include operating system design, resource management, virtualization, distributed and cloud systems, compiler optimization, and fault-tolerant software systems . Advances in system software enable high-performance computing, real-time systems, embedded devices, and secure computing environments , forming the foundation for modern computing infrastructure.

Data Science (DS)		
No.	Technique	Description
1	Data Mining	Extracts patterns, relationships, and knowledge from large datasets.
2	Big Data Analytics	Deals with processing and analyzing massive and complex datasets for insights.
3	Predictive Analytics	Uses historical data and models to forecast future trends or behaviors.
4	Feature Engineering	Focuses on selecting, transforming, and creating meaningful input variables (features) from raw data to improve model performance and insights.
5	Recommendation Systems	Develop algorithms to suggest items or content to users based on preferences or behavior.
6	Data Visualization	Focus on visual representation of data to aid understanding and decision-making.
7	Anomaly Detection	Identifies unusual patterns or outliers in data that may indicate errors, fraud, or important events.
8	Statistical Modeling	Applies mathematical and statistical techniques to represent, analyze, and make inferences from real-world data.
9	Data Governance and Ethics	Studies policies and practices for ensuring data quality, security, and ethical use.

10	Edge and IoT Data Analytics	Analyzes real-time data collected from IoT devices at the edge of networks.
----	-----------------------------	---

Artificial Intelligent (AI)		
No.	Technique	Description
1	Machine Learning (ML)	Develops algorithms that enable computers to learn patterns and make predictions from data.
2	Deep Learning (DL)	Focuses on neural networks with multiple layers to model complex data like images, speech, and text.
3	Natural Language Processing (NLP)	Enables machines to understand, interpret, and generate human language.
4	Computer Vision	Research algorithms that allow machines to interpret and process visual data from images or videos. Focused on visual data interpretation using algorithmic or machine-learning-based methods. Includes detection, classification, segmentation, and feature extraction using models or mathematical operations.
5	Reinforcement Learning (RL)	Studies how agents can learn optimal decision-making through trial-and-error interactions with an environment.
6	Explainable AI (XAI)	Develops methods to make AI models transparent, interpretable, and understandable for humans.
7	Robotics and Autonomous Systems	Focuses on intelligent agents that can sense, plan, and act autonomously in physical or virtual environments.
8	Multi-Agent Systems	Studies systems where multiple intelligent agents interact, cooperate, or compete to achieve goals.
9	AI Ethics and Fairness	Investigates responsible AI practices, addressing bias, fairness, accountability, and transparency.
10	Generative AI	Explores models that can generate new data, such as images, text, or music, from learned patterns.
	Expert Systems & Decision Support Systems (DSS)	Expert Systems and DSS are categorized under intelligent computational systems used to solve domain-specific problems. They rely on rules, structured knowledge, and algorithmic decision

		models to simulate human reasoning and assist users in solving complex decision-making tasks.
	Intelligent Systems	Intelligent Systems are AI-driven solutions capable of reasoning, adapting, or learning from interactions. They include agents, rule-based engines, and systems that simulate human-like cognitive behavior.
	Image Processing	Image/Signal Processing is treated as a project domain for building applications involving audio, image, or sensor-based data processing. The emphasis is on functional output, such as enhancement tools or detection systems.

Note:

Addendum: In case there are other research areas or concerns not mentioned in the listed categories, these shall be initially discussed and settled by the respective **course advisers** to determine their appropriateness and alignment with the program guidelines.

4. Dataset Requirements

4.1. For Data Science specialization, the study must use a dataset containing at least **10,000** records to ensure statistical and analytical validity.

4.2. Software Engineering studies may also require datasets if modeling, training, or testing of components is involved.

- The number of records per dataset depends on the scale of the study: **small-scale** typically use **100–1,000** records for simple models like defect prediction or effort estimation, while **medium-scale** uses **1,000–10,000** records to support more robust models and cross-project studies.

4.3. Artificial Intelligent studies, selecting an appropriate dataset size and structure is essential to ensure meaningful experimentation and proper model evaluation.

- AI research, an ideal dataset size ranges from **1,000 to 50,000 records**, which is sufficient to demonstrate key concepts in modeling, training, and testing. The main goal is to show understanding of the process rather than achieving the highest possible accuracy. It's also important to **split the dataset properly** (around 70% for training, 15% for validation, and 15% for testing) and clearly **document all preprocessing steps**. See appendix B.

4.4. Datasets must be ethically sourced and compliant with data privacy laws (*e.g., Data Privacy Act of 2012*).

5. Group Member Changes

- 5.1. Changes in group membership shall only be permitted on a case-to-case basis and must be approved by both the Thesis Course Adviser.
- 5.2. Valid reasons for member changes include health issues, academic withdrawal, or personal circumstances beyond control.
- 5.3. All approved changes must be documented formally.

6. Faculty Consent and Authorship Agreement

- 6.1. A faculty member may present a thesis title on behalf of a mentee group only with the group's written consent, accompanied by a signed agreement.

7. Title Presentation Guidelines

- 7.1. Each group shall be allotted 30 minutes for title presentation:

10 minutes	Introduction, Presentation of major of the project
20 minutes	Paper Presentation
15-30 minutes	Question and Answer

- 7.2. Presentations must include:

- Proposed Titles
- Objectives of the Study
- Scope and Limitations
- Possible Algorithm/s
- Target Beneficiary

8. Course Grade

- 8.1. Course grade of the students will be computed as follows:

Breakdown	Percentage
Panel Members	60% (total grade of oral defense)
Course Adviser	30% (course requirements)
Project Adviser	10% (mentoring requirements)

9. Title Modification Policy

- 9.1. A group may change their thesis title once after approval, provided it receives endorsement from both the thesis adviser and panel members.
- 9.2. Title change requests must include a written justification and updated title documentation.

10. Roles and Responsibilities

Course Adviser

- Guides students in selecting and refining thesis topics.
- Ensures compliance with institutional thesis guidelines.
- Coordinates scheduling of title presentations and progress checks.

Thesis Adviser (Mentor)

- Provides technical and research guidance throughout the thesis process.
- Reviews and endorses all outputs before submission to the panel for defense.

Panel Members

- Evaluate the technical merit, feasibility, and academic contribution of the study.
- Provide constructive feedback and recommendations during presentations.
- Approve or disapprove title proposals and revisions.

11. Technical Adviser and Investor Consultations (Optional)

- Experienced Technology Project Developers as Speakers in Workshops and Seminars (c/o IALAP)
- Personnel from Different Fields (e.g. IT, Business, Creatives) – c/o IALAP
- Technical Advisers, Panel and Investors (Project adviser and group will work closely with investor to provide project progress)

12. Final Notes

- All submissions and approvals must be properly documented.
- Groups and advisers are encouraged to maintain consistent communication through formal channels (e.g., adviser meetings, thesis forms and letters).
- Submission of scientific papers must include **AI detection verification** while ensuring adherence to Pledge of Honor for MILES Learners which upholds **ethical research conduct and academic integrity standards** at all stages.

APPENDIX A

Machine Learning Algorithm (AI Major)

1. Supervised Learning

- **Description:** Models learn from labeled data to make predictions.
 - **Algorithms / Techniques:**
 - Linear Regression, Logistic Regression
 - Decision Trees, Random Forest
 - Support Vector Machines (SVM)
-

-
- k-Nearest Neighbors (k-NN)
 - Gradient Boosting (XGBoost, LightGBM)

2. Unsupervised Learning

- **Description:** Models find patterns or structure in unlabeled data.
- **Algorithms / Techniques:**
 - k-Means Clustering, Hierarchical Clustering
 - DBSCAN
 - Principal Component Analysis (PCA)
 - Autoencoders

3. Semi-Supervised Learning

- **Description:** Uses a small amount of labeled data with a large amount of unlabeled data.
- **Algorithms / Techniques:**
 - Label Propagation / Label Spreading
 - Semi-Supervised SVM
 - Graph-Based Learning

4. Reinforcement Learning (RL)

- **Description:** Agents learn optimal actions through trial-and-error interactions with an environment.
- **Algorithms / Techniques:**
 - Q-Learning
 - Deep Q-Networks (DQN)
 - Policy Gradient Methods
 - Actor-Critic Methods

5. Ensemble Learning

- **Description:** Combines multiple models to improve predictive performance.
- **Algorithms / Techniques:**
 - Bagging (Random Forest)
 - Boosting (AdaBoost, Gradient Boosting)
 - Stacking

6. Online / Incremental Learning

- **Description:** Models update continuously as new data arrives.
- **Algorithms / Techniques:**
 - Stochastic Gradient Descent (SGD)
 - Online Perceptron
 - Incremental Decision Trees

7. Deep Learning (subset of ML)

- **Description:** Uses multi-layer neural networks to model complex patterns in large datasets.
 - **Algorithms / Techniques:**
 - Convolutional Neural Networks (CNN) – for images
 - Recurrent Neural Networks (RNN) and LSTM – for sequential data
-

-
- Transformers – for language and multimodal tasks
-

APPENDIX B

AI Subfield	Recommended Records (for Students)	Example Tasks	Sample Public Datasets
Machine Learning (General)	1,000 – 10,000	Classification, regression, prediction models	UCI Machine Learning Repository, Kaggle small datasets
Natural Language Processing (NLP)	2,000 – 50,000 text samples	Sentiment analysis, text classification, spam detection	IMDB Reviews, SMS Spam Collection, Yelp Reviews
Computer Vision	1,000 – 50,000 images	Image classification, object detection, handwriting recognition	MNIST (70,000), CIFAR-10 (60,000), Fashion-MNIST
Speech / Audio Processing	1,000 – 10,000 audio clips	Speech command recognition, emotion detection	Google Speech Commands, UrbanSound8K
Reinforcement Learning	~1,000 – 10,000 simulation steps or episodes	Game AI, pathfinding, decision-making	OpenAI Gym (CartPole, MountainCar), FrozenLake
Recommendation Systems	1,000 – 20,000 user-item interactions	Movie, music, or product recommendations	MovieLens 100K, Book-Crossing Dataset
Anomaly Detection / Predictive Maintenance	1,000 – 10,000 records	Detecting unusual patterns or failures	NASA Bearing Dataset, Credit Card Fraud Dataset
Generative AI (basic)	1,000 – 5,000 samples	Simple image or text generation	Small custom datasets, Digit drawing datasets

AI Subfield	Recommended Records (for Students)	Example Tasks	Sample Public Datasets
Explainable AI (XAI) experiments	500 – 5,000 samples	Model explanation and interpretation	Any small ML dataset (e.g., Iris, Wine Quality)